

multiple printing methods like flexographic, dispenser and screen printing, the team hopes to produce a proof-of-concept device by early 2016.

This unique device will harvest and store energy from the waste heat of the engine to keep sensors running for much longer than traditional sensor nodes. It will have a printed thermoelectric generator for generating power from engine waste heat, printed supercapacitor for supplying sufficient duty-cycle power, printed battery for storing harvested energy and long-term power, and an accelerometer for sensing the engine's vibrations to keep a tab on its health.

Ink matters

Ink is another major area of innovation in the world of printed and flexible electronics. One such example is a project at Acreo Swedish ICT to develop ink that can print electronics for different applications, which can be used any time, come rain or shine.

When developing electrochemical components, the key component is electrolyte. If you take the example of traditional batteries or electrolytic condensers, the electrolyte is hermetically sealed in a metal casing. However, when you take the example of a printed display, the electrochemical cell has to function without hermetic encapsulation in all kinds of environments. Additionally, the material must also be non-hazardous, right from manufacture and handling to disposal.

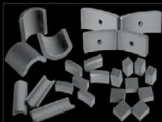
There are other requirements also that electronic ink must meet. It must not dry in the press but must dry quickly once deposited on the substrate. It must spread on the surface at just the right speed so that it can be controlled and also forge well with the substrate. It must be durable, pliable and compatible with other materials that might be used in the product.

Acreo is working to develop an ink, based on polyelectrolytes, which will meet all, or at least most, of these expectations. They are also working with Linköping University to study the curing of ink. In a press report, Dr Mats Sandberg of Acreo explained that, systems for ultraviolet (UV) curing used for graphic printing ink are good enough for bonding pigment but it does not work well enough for functional printing ink. "In order for the electronic printing ink to harden rapidly enough, we need to develop new bonding agent chemistry," he said.

Acreo is working to solve such problems and so are numerous other organisations around the world. Technologists have been working on printed and flexible electronics for a long time now, but it is evident that they are now in a real hurry because there is scope for a number of new applications, which will involve innumerable objects with printed electronics working together with mobile phones and, you guessed it right, the Internet of Things! Indeed, there is a lot of excitement in store for us. ●



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